Mahtab Alizadeh Vandchali

Education

2025 – Present **PhD in Computer Science**, *Rice University*, Houston, TX
2024 – 2025 **M.Sc. in Computer Science**, *Rice University*, Houston, TX
2018 – 2023 **B.Sc. in Mathematics**, *University of Tehran (UT)*, Tehran, Iran

Research Interests

- Optimization for Machine Learning
- Learning Theory
- Low-rank Adaptation (LoRA)
- Theory-guided PEFT

Publications & Preprints

- **Mahtab Alizadeh Vandchali**, Fangshuo (Jasper) Liao, Anastasios Kyrillidis. *One Rank at a Time: Cascading Error Dynamics in Sequential Learning*, May 2025. **[arXiv]**.

Research Experiences

2024 – Present **Graduate Research Assistant**, Computer Science Department, Rice University
Supervisor: Prof. Anastasios Kyrillidis

- **Sequential Rank-1 Linear Deflation and Cascading Error Control**
  Developed a sequential rank-1 deflation framework with a provable error-propagation bound that depends on per-stage inexactness and eigengaps; then validated the bound's prediction (allocate more compute early) via synthetic tests and also LoRA adaptations on MNIST/CIFAR10/CIFAR100, where front-loaded schedules achieved better final reconstruction/training error than equal or late-heavy under the same budget, with LoRA results stable ($\lesssim$ 1.5% std over 5 runs).
- **Nonlinear ReLU Sequential Rank-1 Extension and Concentration-Based Error Control**
  Extended the linear framework to ReLU networks by showing the population gradient is linear up to a bounded remainder and deriving a contraction condition that links stage error to per-stage inexactness and eigengaps; established concentration to translate population results to finite-sample guarantees; validated the recommendation to allocate more compute early via synthetic and feature-based experiments where front-loaded schedules consistently improved final reconstruction/accuracy under the same budget; current focus is decision rules for stage-wise rank and compute allocation, with manuscript in preparation.

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- **LoRA–NTK Linearized Framing Toward Sequential Rank-1 Learning** Formulates LoRA fine-tuning in the NTK regime as a sequential rank-1 learning problem and analyzes how early approximation errors propagate to later components; proposes a comprehensive experimental framework across model architectures and tasks, including variants of sequential training to study trade-offs in computational and parameter efficiency and generalization. Goals: bridge parameter-efficient fine-tuning and the theory of sequential learning to produce practical guidance for compute allocation and, eventually, stage-wise rank criteria; framework and experimental plan in progress.

Professional Activities

- **AAAI 2026** — Program Committee (Main Track).
- **AAAI 2027** — Program Committee (Main Track).

Honors & Awards

- **Graduate Fellowship** — Rice University, CS Department, 2024; **$40,000**.
- **Top 0.3%**, Nationwide University Entrance Exam (Konkoor) among 140,000+ participants.
- **Bronze Medal**, Iran Junior Mathematics Olympiad (Mobtakeran).

Talks & Posters

- "One Rank at a Time: Cascading Error Dynamics in Sequential Learning," *Rice CS Graduate Seminar*, 2025.
- "One Rank at a Time: Cascading Error Dynamics in Sequential Learning," *19th Energy HPC & AI Conference*, 2025 (**Poster**).

Selected Courses

- Optimization: Algorithms, Complexity & Approximations
- Convex Optimization
- Introduction to Deep Learning
- Artificial Intelligence
- Statistical Method
- Data Mining
- Data Structure
- Graph Theory
- Algorithm Design and Analysis

Teaching Experiences

2024 – 2025 **Teaching Assistant**, *Computer Science Department*, Rice University

- Algorithmic Thinking
- IOT Programming Data Analysis

2020 – 2022 **Teaching Assistant**, *Mathematics Department*, University of Tehran

- Linear Algebra
- Graph Theory